

1. Consider a loaded six-sided die that is designed to produce lower-than-average rolls. If we let the random variable X represent the result of a single die roll, what is the expected value of X ? Use the following PMF table and show the calculations you use to get your answer.

x	1	2	3	4	5	6
$P(X = x)$	0.05	0.05	0.10	0.25	0.25	0.30

2. Suppose the weights of newborn babies in a given population are distributed **normally** with a mean of 3.59 kilograms and a standard deviation of 0.54 kilograms. Use built-in R functions to answer the following questions, rounding to 3 significant figures:
 - (a) What is the probability a randomly selected newborn baby weighs less than 2.75 kilograms?
 - (b) What is the probability a randomly selected newborn baby weighs between 3 and 4.25 kilograms?
 - (c) What is the probability a randomly selected newborn baby weighs more than 4.75 kilograms?
 - (d) Below what weight in kilograms are the lightest 25% of newborn babies?
3. Management at a store is monitoring how frequently customers enter. They determine that customer frequency can be modeled by the exponential distribution, with a rate of $\lambda = 1/7$. Use built-in R functions to answer the following questions, rounding to 3 significant figures.

Hint: The exponential distribution has a “rate” parameter in RStudio that you will set equal to the provided λ value.

 - (a) What is the probability a new customer enters within the next 12 minutes after the previous customer?
 - (b) Within how many minutes can you expect 90% of new customers to arrive after the previous customer?
4. Assume $X \sim U(a, b)$ is a continuous random variable. Calculate the **expected value** of the random variable X . Include your work and calculations. You may want to submit your written work in a separate file or picture, though showing your work in your knitted PDF file (as LaTeX or an embedded image) is fine as well. Recall, for the uniform (continuous) random variable,

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for all other } x \end{cases}$$